



# Paolo De Angelis, Ph.D.

PostDoc in Energetics Engineer

paolodeangelis.github.io

## Contact

Largo Francia 111, 10138  
Torino (TO) - Italy

paolo.deangelis@polito.it  
paolo1193@gmail.com

(+39) 348 6604548

paolo-de-angelis

paolodeangelis

## About

11/11/1993

Italian

Male (he/him/his)

## Languages

Italian native

English upper intermediate

## Web presence

0000-0003-1866-2988

jeSYLUUAAAAJ

57212316557

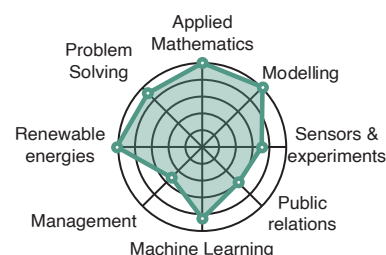
Paolo-De-Angelis

dap.93

paolo.de.angelis

Paolo1193

## Professional interests



I am a Postdoctoral Researcher specializing in atomistic modeling for energy storage materials, with expertise spanning thermodynamics, computational chemistry, statistical physics, and machine learning (ML). My research focuses on Li-ion battery (LIB) interface reactions (SEI) using multi-scale approaches. My technical toolkit includes the parametrization of interatomic potentials, from classical to ML-based (MLIP), the use of enhanced sampling methods, and high-throughput DFT and MD simulations on HPC. Beyond technical work, I collaborate with multiple research groups, mentor junior researchers, co-authoring grant proposals, and manage EU project. My short-term goals are to secure a prestigious postdoctoral fellowship to advance my expertise in ML-driven materials study, discovery, and design. In the long term, I aim to establish a robust research profile to apply for funding, such as MSCA, ERC, and Horizon EU, and ultimately securing an academic position.

## Education

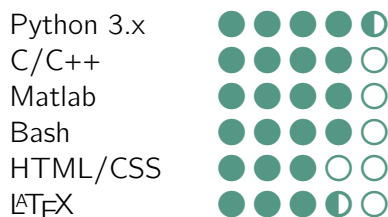
- 2018 - 2022 **PhD in Energetics** Politecnico di Torino, Italy  
PhD defended on November 2, 2022. Thesis titled "Reactive and Non-Reactive Interface Modelling for Energy Materials".   
Language: *English* Mark: 110L/110
- 2015 - 2017 **MSc in Mechanical Engineering** Politecnico di Torino, Italy  
University 2nd cycle degree awarded on December 7, 2017. Thesis titled "Simulation of the ensemble and cycle-resolved combustion processes in an NG-H2 engine".   
Language: *English* Mark: 110/110
- 2012 - 2015 **BSc in Mechanical Engineering** Politecnico di Torino, Italy  
University 1st cycle degree awarded on 9 December 2015. Thesis titled "Quaternion application to 3D roto-translation on simulative environment".   
Language: *Italian* Mark: 107/110
- 2007 - 2012 **Scientific Diploma** Scientific High School 'B. Rosetti', Italy  
Italian secondary school diploma. (Discussed essay: )   
Language: *Italian* Mark: 84/100

## Experience

- 2023 - now **Affiliated Researcher** Italian Institute of Technology, Italy  
Close collaboration with Prof. Parrinello's modeling group to develop MLIP for investigating lithium-ion mass transport dynamics in LiF inorganic salt incorporating ML tools and enhanced sampling techniques for Free Energy analysis.
- 2022 - now **PostDoc on LIB Modeling** Politecnico di Torino, Italy  
Postdoctoral researcher in Prof. Chiavazzo's and Prof. Asinari's research group . Developing Reactive FF to investigate the Solid Electrolyte Interface (SEI) in LIB and exploring ML and Graph Neural Networks (GNN) for materials discovery.
- 2020 - 2020 **Visiting at UIC** University of Illinois at Chicago, USA  
Visiting researcher at the University of Illinois in Chicago (UIC) Mechanical Department, where I developed a multiphysics model of a LIB to train a Deep Neural Network for predicting thermal runaway. (ECS PRIME 2020 Presentation )

## Coding Skills

Proficiency in programming language usage



## Softwares

Short list of know software

### Operating Systems



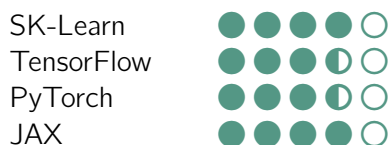
### CFD



### ATOMISTIC



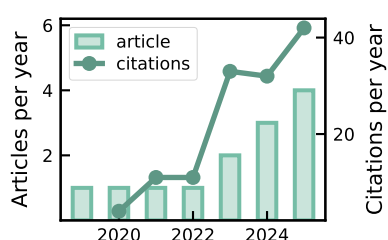
### ML



## Research Output

citations: 135 (Scholar)

h-index: 7 (Scholar)



## Places Lived



## Teaching

- 2024 - now **Machine Learning for energy applications** [01SJFND](#)
- Teaching assistant for 12h lab sessions (in English).
- 2023 - now **Aerosol technology and air quality** [01USDND](#)
- Teaching assistant for 9h lectures (in English).
- 2023 - 2024 **Sistemi a combustione** [06GKYND](#)
- (EN: Combustion systems) Teaching assistant for 24h lab sessions (in Italian).
- 2021 - now **Applicazioni energetiche dei materiali** [02QZSND](#)
- (EN: Advance energy application materials) Teaching assistant for 1.5h lectures and 3h lab sessions (in Italian).
- 2021 - now **Energy storage** [01TVHND](#)
- Teaching assistant for 4.5h lectures (in English).

## Publications

† Equal contribution (co-first) \* Corresponding author

- [1] **P. De Angelis**<sup>\*</sup> *et al.*, *Exploring Lithium Diffusion in LiF with Machine Learning Potentials: From Point Defects to Collective Ring Diffusion*, (Submitted to "NPJ Comp. Mat."), 2025. DOI: [10.26434/chemrxiv-2025-fv0z1](#).
- [2] **P. De Angelis**, G. Barletta, G. Trezza, P. Asinari, and E. Chiavazzo, "Energy-GNoME: A living database of selected materials for energy applications," *en, Energy and AI*, 2025. DOI: [10.1016/j.egyai.2025.100605](#).
- [3] P. Asinari *et al.*, *Notes on Quantum Computing for Thermal Science*, 2025. DOI: [10.48550/ARXIV.2503.19109](#).
- [4] A. Muhammad<sup>†</sup>, C. Valero<sup>†</sup>, **P. De Angelis**<sup>†</sup>, *et al.*, "From nanoscale to printed products: Multiscale modeling and experimental characterization of graphene-enhanced polylactic acid composites for 3D printing," *en, Composites Part B: Engineering*, 2025. DOI: [10.1016/j.compositesb.2025.112354](#).
- [5] M. Casini, **P. De Angelis**, *et al.*, "Machine Learning and image analysis towards improved energy management in Industry 4.0: A practical case study on quality control," *en, Energy Efficiency*, 2024. DOI: [10.1007/s12053-024-10228-7](#).
- [6] M. Casini<sup>†</sup>, **P. De Angelis**<sup>†</sup>, *et al.*, "Current trends on the use of deep learning methods for image analysis in energy applications," *en, Energy and AI*, 2024. DOI: [10.1016/j.egyai.2023.100330](#).
- [7] **P. De Angelis**<sup>\*</sup>, R. Cappabianca, M. Fasano, P. Asinari, and E. Chiavazzo, "Enhancing ReaxFF for molecular dynamics simulations of lithium-ion batteries: An interactive reparameterization protocol," *en, Scientific Reports*, 2024. DOI: [10.1038/s41598-023-50978-5](#).
- [8] A. Andreozzi *et al.*, "Heat Transfer and Thermal Energy Storage Enhancement by Foams and Nanoparticles," *en, Energies*, 2023. DOI: [10.3390/en16217421](#).
- [9] R. Cappabianca, **P. De Angelis**, M. Fasano, E. Chiavazzo, and P. Asinari, "An Overview on Transport Phenomena within Solid Electrolyte Interphase and Their Impact on the Performance and Durability of Lithium-Ion Batteries," *en, Energies*, 2023. DOI: [10.3390/en16135003](#).
- [10] R. Cappabianca<sup>†</sup>, **P. De Angelis**<sup>†</sup>, A. Cardellini, E. Chiavazzo, and P. Asinari, "Assembling Biocompatible Polymers on Gold Nanoparticles: Toward a Rational Design of Particle Shape by Molecular Dynamics," *en, ACS Omega*, 2022. DOI: [10.1021/acsomega.2c05218](#).
- [11] **P. De Angelis** *et al.*, "Data-driven appraisal of renewable energy potentials for sustainable freshwater production in Africa," *en, Renewable and Sustainable Energy Reviews*, 2021. DOI: [10.1016/j.rser.2021.111414](#).

## Hobbies & Interests

Painting & Drawing, Music, Reading, Volunteer Work, Open Source Software, Art Museums, Arduino Projects, Hackathons & Problem Solving, Emerging Technologies, Political Debates, Chess, Cycling, Hiking, Board Games, Cooking, and Traveling.

- [12] **P. De Angelis**, A. Cardellini, and P. Asinari, "Exploring the Free Energy Landscape To Predict the Surfactant Adsorption Isotherm at the Nanoparticle-Water Interface," en, *ACS Central Science*, 2019. DOI: 10.1021/acscentsci.9b00773.

## Conferences & workshops

|         |                                                                                                       |                         |
|---------|-------------------------------------------------------------------------------------------------------|-------------------------|
| 12/2025 | <b>Materials Research Society (MRS)</b>                                                               | Boston, USA             |
|         | Oral presentation (15') and Poster presentation.                                                      |                         |
| 09/2025 | <b>Nanoinnovation</b>                                                                                 | Roma, Italy             |
|         | Invited presentation (15') and sessions Chairman.                                                     |                         |
| 09/2025 | <b>ICTP - Conference on Frontiers in Atomistic Simulations: from Physics to Chemistry and Biology</b> | Trieste, Italy          |
|         | Invited poster presentation.                                                                          |                         |
| 10/2024 | <b>CECAM - Advances in catalytic reactivity simulations under operando conditions</b>                 | Varigotti, Italy        |
|         | Poster.                                                                                               |                         |
| 09/2024 | <b>European Materials Research Society (E-MRS)</b>                                                    | Warsaw, Poland          |
|         | Invited presentation (30') and sessions Chairman.                                                     |                         |
| 09/2024 | <b>Nanoinnovation</b>                                                                                 | Roma, Italy             |
|         | Invited presentation (15').                                                                           |                         |
| 09/2024 | <b>Symposium on Beyond Li-Ion Batteries - BeLi24</b>                                                  | Padova, Italy           |
|         | Presentation (15').                                                                                   |                         |
| 06/2024 | <b>23<sup>rd</sup> Symposium on Thermophysical Properties</b>                                         | Boulder, USA            |
|         | Presentation (15').                                                                                   |                         |
| 06/2024 | <b>41<sup>st</sup> UIT International Heat Transfer Conference</b>                                     | Napoli, Italy           |
|         | Presentation (15').                                                                                   |                         |
| 05/2022 | <b>Materials Research Society (MRS)</b>                                                               | Honolulu, USA           |
|         | Presentation (15').                                                                                   |                         |
| 06/2021 | <b>15<sup>th</sup> AIGE conference</b>                                                                | Ancona, Italy           |
|         | Presentation (15').                                                                                   |                         |
| 10/2020 | <b>ECS PRiME</b>                                                                                      | (Virtual) Honolulu, USA |
|         | Presentation (15').                                                                                   |                         |
| 03/2020 | <b>American Physical Society (APS)</b>                                                                | (Virtual) Denver, USA   |
|         | Presentation (15').                                                                                   |                         |
| 06/2019 | <b>CECAM - Materials Simulation Days</b>                                                              | Mainz, Germany          |
|         | Poster.                                                                                               |                         |

## References

### Professor **Pietro Asinari**

📍 *INRiM, 10135, Torino TO, Italy*

📍 *Politecnico di Torino, 10129, Torino TO, Italy*

✉ [p.asinari@inrim.it](mailto:p.asinari@inrim.it)

✉ [pietro.asinari@polito.it](mailto:pietro.asinari@polito.it)

☎ (+39) 011 3919310

### Professor **Eliodoro Chiavazzo**

📍 *Politecnico di Torino, 10129, Torino TO, Italy*

✉ [eliodoro.chiavazzo@polito.it](mailto:eliodoro.chiavazzo@polito.it)

☎ (+39) 011 0904557